Supplementary Information (Online Resource 4)

Article title: How Teams Decide Human Oversight in AI-Powered Systems: A Mixed-Methods Study

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Ontology-based questionnaire

*Human-oversight-related questions are highlighted in blue

#	Objectives	RE stage	Stakeholders
1	1.1 - What was the business need or motivation that caused the initiation of the AI-driven project? 1.2 - What are the factors that influence the decision to use AI over conventional methods?	1.3 - What technical, financial, and human resources are required for the successful integration and deployment of the AI solution? 1.4 – How can we assess the readiness and feasibility of AI solutions for specific tasks and contexts? 1.5 – What are the initial stages of assessing the system's readiness for AI integration? 1.6 - What are the effective methods, procedures and tools for developing and managing the complete requirements for AI-powered systems? 1.7 - What are the tools and techniques that can be used for verifying and validating the requirements of AI systems? 1.8 - What are the testing methods specifically required for AI-powered systems? 1.9 - What are the primary causes of requirement changes in AI-powered systems, and what are the best practices for managing these changes?	1.10 – What are the roles and responsibilities of different stakeholders in the process of developing requirements of AI systems, both functional, quality, performance and design? 1.11 – How can we ensure and encourage their effective participation and collaboration, ensuring that their feedback is properly considered and integrated? 1.12 – What are the challenges and best practices for communicating and negotiating the requirements with different stakeholders? 1.13 – How can we surmount the diverse attitudes, perceptions, and understanding of AI among stakeholders? 1.14 – What are the challenges and best practices for communicating and negotiating the requirements with different stakeholders?

#	Requirement category	Objectives	RE stage	tensions and trade-offs	Stakeholders
2	Human-system interaction (Automation vs. human oversight)	2.1 - What is the anticipated level of human involvement in the system operation? 2.2 - What are the primary justifications for human intervention within the system operation?	Addressed in other questions*	2.3 - How should the system balance automation and user control to maintain usability? 2.4 - How can we provide human oversight while maintaining system efficiency? 2.5 - What ethical considerations should be accounted for in human-system interactions?	Users & Domain Experts

				2.6 - How does reducing human oversight impact user trust, and in which scenarios should human intervention be mandatory to maintain trust?	
3	Trust\trustworthiness	3.1 – What are the key factors that contribute to user trust in the systems? 3.2 – How do these factors vary across different contexts and applications?	3.3 - What are the best practices for evaluating and measuring user trust in AI systems?	Addressed in other questions*	Users
4	Ethics	Addressed in other questions*	4.1 - How can we ensure that ethical considerations are effectively integrated into the development and deployment of AI systems? 4.2 – What validation methods are used to confirm that ethical considerations are effectively integrated into the system? 4.3 - How can we ensure that ethical requirements are managed along the system life-cycle? 4.4 - What measures can be taken to address violations of ethics within the system?	4.5 - What are the trade-offs and constraints that ethical considerations impose on other system requirements, particularly performance and usability?	Ethical experts
5	Fairness	5.1 - What legal, ethical, and social considerations should guide the fairness requirements in the system? 5.2 - What are the key fairness risks specific to the system, and how can they be mitigated?	5.3 - How can fairness be tested in outputs, and what validation methods should be applied?	5.4 - How does ensuring fairness impact system performance? 5.5 – How does ensuring fairness affect quality aspects such as transparency, privacy, security, and system robustness, and what strategies can be used to balance these requirements effectively?	Ethical experts and regulators
6	Privacy	6.1 - How can existing privacy frameworks and regulations, such as GDPR, be adapted and extended to effectively govern the use of AI systems?	Addressed in other questions*	6.2 – How do privacy requirements influence system design, performance, and other quality attributes such as fairness and security? 6.3 – How can the effectiveness of privacy be validated to ensure compliance with regulatory standards?	Ethical experts and regulators
7	Safety	7.1 - What are the key safety objectives that the system must achieve to prevent harm and ensure operational reliability? 7.2 - How should risk assessment methodologies be applied to define safety requirements?	7.3 - How can the safety functions be validated to ensure they meet safety standards?	7.4 - What level of human oversight should be included? 7.5 – How do safety-related AI decisions impact system performance and other quality attributes?	Safety experts and regulators
8	Security	8.1 – What vulnerabilities arise from using AI models? 8.2 - What specific measures can be implemented to address potential vulnerabilities and threats?	8.3 - How can security be assessed across different stages of development?	8.4 – How can the system ensure security while maintaining explainable? 8.5 - What usability challenges arise from security mechanisms? 8.6 - Does increasing security measures (e.g., restricted access, encrypted logs) reduce the	Security experts

				system's responsiveness or efficiency? 8.7 - How can the system be made resilient against adversarial attacks while maintaining flexibility and robustness? 8.8 - Can strict security controls lead to ethical concerns such as user profiling or discrimination?	
9	Sustainability	9.1 - What sustainability regulations apply to the system's domain? 9.2 - How will the system measure and report compliance with sustainability metrics?	Addressed in other questions*	9.3 – How can we optimize system performance while reducing computational consumption and minimizing environmental impact?	Sustainability expert
10	Explainability & transparency	10.1 - How should explainability requirements be defined for different stakeholders, such as developers, users, and regulatory bodies, to ensure appropriate levels? 10.2 – What methods are used for explainability? 10.3 - Could the explainability requirement issue be a "show stopper" for the decision not to use AI for a specific task?	10.4 - What metrics and validation methods should be used to evaluate the effectiveness of AI explanations?	10.5 – How does the need for explainability impact the selection of models and the decision to opt for traditional methods? 10.6 – How does ensuring transparency impact system performance and quality requirements robustness, privacy and security? 10.7 – What are the most effective strategies for balancing and optimizing trade-offs?	AI experts and ethical experts
11	Robustness	11.1 - How can we design AI systems that are both resilience to noise and adversarial attacks but also flexible enough to evolve with new data?	11.2 - What validation methods should be used to assess the robustness of the system, ensuring it performs reliably across different conditions, inputs, and adversarial scenarios?	11.3 - How do we decide whether to prioritize highly specified models (for accuracy) or generalized models (for adaptability and evolvement to new real-world data)? 11.4 - How do we balance dataset diversity to improve generalization while minimizing instability and noise in the model? 11.5 – How can we balance safety, fairness and privacy requirements with system robustness.	AI experts & data scientists
12	Usability	12.1 - What are the key usability challenges specific to AI-powered interactions, and how should they be addressed?	12.2 - How should usability metrics be defined and measured?	12.3 - What trade-offs exist between system explainability and usability? 12.4 - How can usability be maintained while ensuring privacy, security, and compliance with ethical guidelines?	Users
13	Trainability	13.1 - What retraining policies should be established to maintain AI performance over time? 13.2 – What tools, methods, and technologies are required to perform trainability?	Addressed in other questions	13.3 - How do resource constraints affect the frequency and feasibility of AI model retraining?	AI experts & data scientists

14	Model	14.1 - What challenges and best practices should be considered when selecting the most suitable AI model?	14.2 – What is the projected evolution of the model, and how can this evolution be managed? 14.3 - What mechanisms should be in place to monitor, detect, and mitigate AI model deviations or failures throughout the system lifecycle?	14.4 – How does data availability and qualities impact model selection?	AI experts & data scientists
15	Data	15.1 - What are the most critical data qualities, and what data preparation steps are necessary for the successful implementation of AI capabilities?	15.2 - What are the requirements for measuring and evaluating data qualities and data drifts along the system life cycle? 15.3 – What are the appropriate timing and triggers for conducting this evaluation?	15.4 – What potential risks are associated with the selection of specific features in the data? 15.5 - What data and model requirements are necessary to prevent biases that could affect the system's fairness? 15.6 – How does noisy data impact system performance, and what strategies can be implemented to mitigate its effects?	Data scientists
16	Computation resources	16.1 - What are the computational resource limitations encountered when developing and deploying AI systems?	16.2 - How can the availability and adequacy of computational resources be validated to ensure the efficient operation of the AI system, both for training and real-time inference?	16.3 - How does the availability of computing power impact model selection and optimization? 16.4 What optimization techniques are used to ensure AI models operate efficiently within limited resources?	AI experts & IT architects
17	Performance	17.1 - What are the key performance indicators (KPIs) for the AI system?	17.2 - What validation techniques should be used to assess performance along the system life-cycle? 17.3 - What monitoring tools should be implemented to continuously track AI performance?	17.4 - Which key quality requirements conflict with system performance, and what are the acceptable trade-offs to balance them? 17. 5 - What constraints does the computational environment impose on AI performance, and how do they impact system efficiency and accuracy?	Users & AI experts